

Mission Antyodaya 2020 Cluster Analysis

What 6 Lakh Indian Villages Tell Us About Rural Development

Amit Kumar, CoRE Stack | Research supervision: Prof. Aaditeshwar Seth, IIT Delhi

The Mission Antyodaya 2020 survey records rural development through a large number of variables covering public services, institutions, infrastructure, livelihoods, agriculture, health, water, sanitation, and connectivity. Read separately, these variables are difficult to compare and do not readily show how different aspects of a village interact. Our analysis reduces this complexity to a smaller set of indices while retaining the component features needed for diagnosis and further analysis.

Data Preparation

Of the 198 parameters available across 32 state and union-territory files, 121 were selected for analysis. They were transformed into 64 features using normalization, binary availability measures, inversion, ratios, ordinal mappings, and composite scores. All features were oriented so that a higher value represents more of the measured positive condition. Examples include piped-water coverage, hours of domestic electricity, child nutrition and development, land utilization, agricultural risk support, and access to markets.

The 64 features were organized into 21 categories. Features were classified as Low, Medium, and High where sufficient variation existed; binary indicators used Low and High. Within each category, these classes were scored (Low = 0, Medium = 0.5, High = 1) and combined with equal weight. MiniBatch K-means was then used to classify the resulting category index. The output retains both the category summary and its underlying feature values and classes.

Reading the Current Files

The analytical GeoPackage and GeoServer layer retain the 64 feature values so that the method can still be inspected. The generated Excel sheet and village Know Your Landscape output use a smaller contract: 21 category classes, 21 category values, and 109 raw survey fields. No `*_feat_*` column is used in that report flow. Start with a category class and value, then check the raw survey fields placed with that village. Use the full analytical layer only when the intermediate feature construction itself needs to be examined.

Interpreting the Results

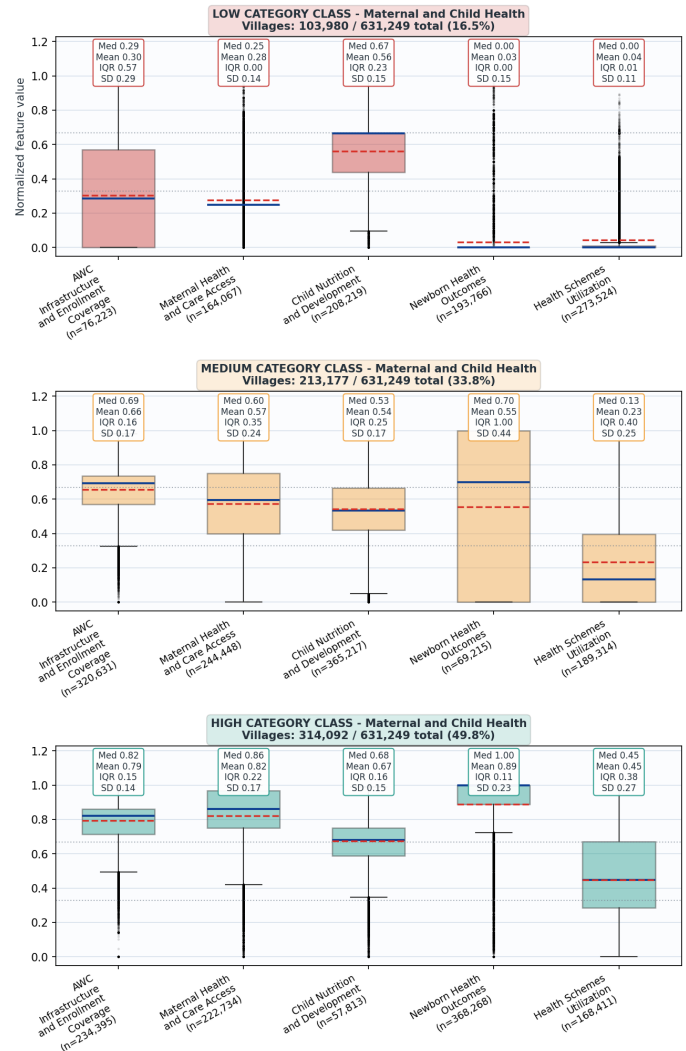
Figure 1 shows how a category index should be opened up and examined. Each panel contains the villages assigned to one Maternal and Child Health class: Low, Medium, or High. Within a panel, each box represents one of the five component features. The blue line is the median, the red dashed line is the mean, the box covers the middle half of observed values, and the whiskers and points show the wider spread.

Read the plots in two directions: follow one feature across the Low, Medium, and High panels, and then compare all features within one panel. This separates differences between classes from internal gaps within a class.

The High class illustrates the point. Its median is 0.82 for Anganwadi infrastructure and enrolment, 0.86 for maternal health and care access, 0.68 for child nutrition and development, 1.00 for newborn health outcomes, and 0.45 for health-scheme utilization. Thus, “High” is a summary of the overall configuration; it does not mean that every component is uniformly high. The lower value and wider spread for health-scheme uti-

lization identify an internal gap that would be hidden if only the category label were read.

Maternal and Child Health



Blue line = median | Red dashed line = mean | Box = Q1-Q3 | Whiskers = 1.5x IQR | n = villages in that feature class

Figure 1: Maternal and Child Health feature distributions across the three category classes.

Across classes, Anganwadi infrastructure and enrolment rises from a median of 0.29 in Low to 0.69 in Medium and 0.82 in High; maternal health and care access rises from 0.25 to 0.60 and 0.86; and newborn health outcomes rises from 0.00 to 0.70 and 1.00. The pattern is not uniformly monotonic: median Child Nutrition and Development is 0.67 in Low, 0.53 in Medium, and 0.68 in High. Health-scheme utilization improves across classes but remains comparatively weak even in High. Class membership therefore reflects the joint configuration of features, not a fixed threshold that every feature crosses.

The category label is a starting point. It should be read with component distributions, sample sizes, and feature definitions.

What it Unlocks

Category and feature classes can be used as filtering and sorting indicators. Their combinations identify patterns that merit closer study and make the reason for selecting a group of villages transparent. Four examples from the national results illustrate the approach.

Water-security risk: 111,658 villages (17.7%).

Pattern: High Farm Employment + Medium/High Land Cultivation + Low Irrigation and Watershed.

Cultivation and dependence on farming are both substantial, while the recorded water-security system is weak. The pattern identifies villages where rainfall exposure, cropping intensity, crop failure, and scope for diversification should be examined.

Agricultural dependence without protective systems: 84,118 villages (13.3%).

Pattern: High Farm Employment + Medium/High Land Cultivation + Low Agriculture Support Services + Low Agricultural Markets + Low Financial Inclusion.

These villages combine agricultural dependence with weak recorded access to support services, finance, storage, and markets. The configuration identifies conditions under which productive effort may not translate into stable producer income.

Collective capacity without enterprise conversion: 42,744 villages.

Pattern: High SHG Federation + Low SHG Credit + Low FPO/PACS Strength + Low Agricultural Markets.

A collective base is present, but links to formal credit, producer institutions, and markets are weak. The filter asks whether SHG structures could support producer groups, enterprise activity, or market aggregation.

Connectivity without local economic conversion: 34,931 villages.

Pattern: High Road Connectivity + High Energy Access + Low Financial Inclusion + Low Agricultural Markets + Low Cottage and Traditional Industry.

Roads and energy are present, while the recorded financial, market, and enterprise systems remain weak. The filter helps locate places where physical access has not yet been accompanied by local economic activity.

The same logic can identify opportunity patterns. High Common Pastures with Low Livestock and Veterinary Services may indicate a resource base without adequate animal-health or market support. High Fisheries with weak finance, roads, or cold-chain access may indicate an existing livelihood constrained by a missing service layer.

Spatial Analysis

Figure 2 is a sample map demonstrating variation in Water and Sanitation metrics across Maharashtra at district level. Village-level category values have been aggregated to district means and compared with the Maharashtra mean of 0.450. Red districts fall below that mean and blue districts rise above it; hatched areas (Mumbai) have insufficient mapped survey coverage. In this example, district means range from 0.316 in

Gadchiroli to 0.641 in Kolhapur.

Maharashtra: Water & Sanitation

District mean village score. Directional plane explains 39.8% of district variation; higher toward southwest, lower toward northeast.

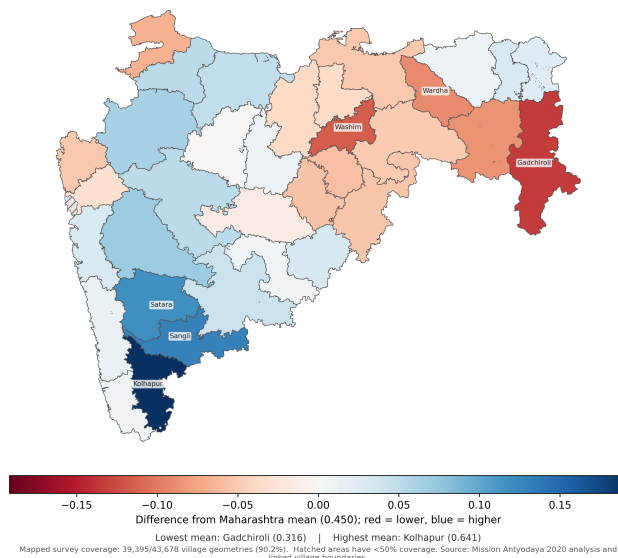


Figure 2: District mean Water and Sanitation scores in Maharashtra.

A district mean can locate a broad geographic pattern, but it can conceal substantial variation among its villages. Users should inspect the village values within the district, open the Water and Sanitation category into its component features, verify data coverage, and compare the result with current administrative and field information.

The same procedure can be repeated across India for any of the 21 categories or 64 features. Depending on the question and the available boundary data, results can be viewed at village level or summarized at sub-district, district, state, or national scale.

Limitations

The classes are relative groupings in the 2020 data, not service standards, administrative rankings, causal findings, or programme recommendations. Missing values, uneven reporting, aggregation, ecology, and local economic history can affect interpretation. Results should be verified against feature definitions, raw and current data, and field conditions.

Conclusion

The value of the framework lies in moving between levels of evidence: from a category class, to its component features, to a cross-category pattern, and then to a spatial and field question. Used in this way, the analysis can help researchers, public institutions, civil society, and village communities identify risk and opportunity patterns. It narrows the field of enquiry and makes the assumptions behind prioritization visible.

References and data: Mission Antyodaya 2020 Village Cluster Analysis Report; village-level normalized values and cluster labels; pan-India GeoPackage; Mission Antyodaya 2020 raw data files; GEE Antyodaya 2020 Explorer. Data preparation and analytical assistance: Ananya Singh, IIT Delhi.